

Regression:

t_i i th target value

x_i i th input value

$y(x)$ predicted value for x

$\mathbf{w} = (w_0, w_1, \dots, w_M)^T$ vector of model parameters

$\boldsymbol{\phi}(x) = (\phi_0(x), \phi_1(x), \dots, \phi_M(x))^T$ vector of basis functions

$y(x) = \mathbf{w}^T \boldsymbol{\phi}(x) = w_0 \phi_0 + \dots + w_M \phi_M$

Classification:

X_i i th feature

w_i i th weight

b model bias

$y(X_1, X_2) = w_1 X_1 + w_2 X_2 + b$ model prediction for point (X_1, X_2)

KKT conditions:

$\sum \alpha_n t_n = 0$ where α_n is the n th Lagrange multiplier

$\alpha_n (1 - y_n t_n) = 0$

Margin = $1/||\mathbf{w}'||$

Clustering:

K-means kernels:

- Polynomial:

$$\kappa(x, y) = (x^T y)^p$$

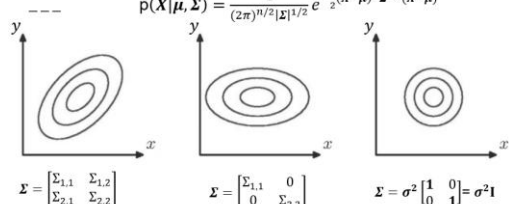
- Gaussian:

$$\kappa(x, y) = \exp(-\lambda |x - y|^2)$$

x, y : feature representations of data points

p : degree of the polynomial kernel

λ : Positive scaling parameter that controls the shape of the Gaussian kernel.

Mixture models – Gaussian Distribution in 2D	
--- $p(\mathbf{X} \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{n/2} \boldsymbol{\Sigma} ^{1/2}} e^{-\frac{1}{2}(\mathbf{X}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu})}$  $\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{1,1} & \Sigma_{1,2} \\ \Sigma_{2,1} & \Sigma_{2,2} \end{bmatrix}$ $\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{1,1} & 0 \\ 0 & \Sigma_{2,2} \end{bmatrix}$ $\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \sigma^2 \mathbf{I}$	--- Gaussian mixture model - Assume component distributions are Gaussians with diagonal covariance: $p(\mathbf{x}_n z_{nk} = 1, \boldsymbol{\mu}_k, \sigma_k^2) = \mathcal{N}(\boldsymbol{\mu}_k, \sigma_k^2 \mathbf{I})$ - We need to be able to estimate the prior of assignment. Let $\pi_k = p(z_{nk} = 1 \Delta)$ - We also want to estimate the probability to assign data to each component $q_{nk} = \frac{\pi_k p(\mathbf{x}_n z_{nk} = 1, \boldsymbol{\mu}_k, \sigma_k^2)}{\sum_{j=1}^K \pi_j p(\mathbf{x}_n z_{nj} = 1, \boldsymbol{\mu}_j, \sigma_j^2)}$

Mixture model optimisation – the Expectation-Maximization (EM) algorithm

- Following optimisation algorithm:
 1. Guess μ_k, σ_k^2, π_k
 2. **(E)**xpectation-step: Compute q_{nk}
 3. **(M)**aximization-step: Update μ_k, σ_k^2, π_k
 4. Return to 2 unless parameters are unchanged.
- Guaranteed to converge to a local maximum of the lower bound.

Mixture model optimisation – the Expectation-Maximization (EM) algorithm

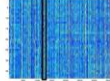
- Following optimisation algorithm:
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$$\hat{N}_k = \sum_{i=1}^N q_{ik} \quad \hat{\mu}_k = \frac{1}{\hat{N}_k} \sum_{i=1}^N q_{ik} x_i \quad \hat{\Sigma}_k = \frac{1}{\hat{N}_k} \sum_{i=1}^N q_{ik} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

Projection:

Finding a subset – example

- Take one feature – N values.



- Some values from objects in class 1, some from class 0.
- Split them based on class and compute μ and σ^2 for each class.
- Compute s for each feature:

$$s = \frac{|\mu_1 - \mu_0|}{\sigma_0^2 + \sigma_1^2}$$

- Keep features with high s .

Principal Components Analysis

- Principal Components Analysis (PCA) is a method for choosing $\mathbf{W}$.
- It finds the columns of $\mathbf{W}$ one at a time (define the m th column as $\mathbf{w}_m$).
 - Each $D \times 1$ column defines one new dimension.
- Consider one of the new dimensions (columns of $\mathbf{Z}$):

$$\mathbf{z}_m = \mathbf{X}\mathbf{w}_m$$

- PCA chooses $\mathbf{w}_m$ to maximise the variance of $\mathbf{z}_m$

$$\frac{1}{N} \sum_{n=1}^N (z_{mn} - \mu_m)^2, \quad \mu_m = \frac{1}{N} \sum_{n=1}^N z_{mn}$$